From Credit Terms to Shelf Prices: Financial Frictions and the Margins of Price Adjustment

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Motivation: Do financially constrained firms raise or cut prices?

- If tighter credit *raises* prices, it works **against** rate hikes (Barth–Ramey 2001)
- Real side: cash raised through prices need not come from cutting **investment** or employment – not today
- How do financially constrained firms respond?
  - **Harvest**: *raise* prices, pulling cash from installed customers – Gilchrist et al. 2017, Renkin–Züllig 2024, Berardi 2024
  - **Liquidate**: *cut* prices, generating cash from volume – Kim 2021
- Did tighter credit add to the 2021–23 inflation surge or dampen it?

▸ Related literature

How do banks regulate credit? Rationing and Pricing

- **Rationing**: banks become less willing to lend – the *quantity* of credit ⇒ credit **access**
- **Pricing**: banks charge a higher interest rate – the *price* of credit ⇒ credit **cost**
- These are not the same. From survey data we observe (2018–2024 panel):

	What the firm reports about its bank			
Outcome reported	No tightening	Cost only	Access only	Both
Financially constrained	6.6%	6.3%	23.9%	31.5%
of which: application rejected	2.4%	0.0%	8.0%	11.3%

▸ Credit-outcome taxonomy

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Repricing leaves borrowing intact, rationing makes the constraint bind

Research question

Do credit-constrained firms **cut** prices to generate liquidity from sales, or **raise** them to harvest cash flow from their customer base?

Do they adjust through the size or the frequency of price changes?

This paper

- **Data**: scanner prices (PRISMA/IRi) × producers' balance sheets (Orbis) × each producer's own credit terms at a dated interview (SAFE)
- **Evidence**: new stylized facts on price setting, leveraging the interviews and the margins of price adjustment
- **Model**: state-dependent pricing with a customer base. Rationing makes cash **scarce**, a higher loan rate makes cash **expensive** – estimated on the four margins

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Main idea: rationing creates cash demand, high interest rates do not

Idea: cash demand pushes prices down in small steps over time, expensive cash pushes the price level up

Rationing (access ↓): cash demand

- Firms need cash ⇒ cut prices to sell more (*liquidate*)
- Large price changes lose customers ⇒ cuts frequent but *shallow*, increases rare and small

⇒ Prices change often, the **level barely moves**

Pricing (cost ↑): high interest rates

- Higher interest rate ⇒ producing costs more ⇒ higher target price
- Passed on mainly by *cutting less often*

⇒ The **price level rises**

Preview of results

- **Higher interest rates raise inflation**: +1.4pp/month, via *fewer cuts*
- **Rationing lowers inflation**: −1.6pp/month – increases rarer (−9.5pp) and smaller, cuts more frequent (+15pp) and shallower
- **Liquidity sets the direction**: rationed firms cut rather than raise
- **Customer base sets the size**: cuts are shallow, while cash need alone would make them deeper
- **Heterogeneity** in responses to access by leverage and liquidity: strongest in cash-poor firms (−2.5 vs. −1.4pp/month in cash-rich) and the most levered
- **2021–23**: tight credit adds $\approx$ +1.0pp cumulative inflation to the covered grocery basket – essentially all via higher interest rates

Data

Three data sets, one monthly panel

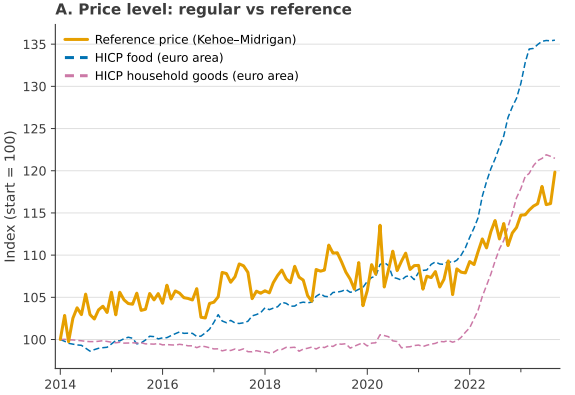
- **PRISMA/IRi scanner data**: weekly product–store prices and quantities, 2013–2023, Germany, France, Italy, Netherlands, aggregated to marketing categories
- **Orbis** (Bureau van Dijk): annual producer balance sheets – size, profitability, liquidity, leverage, ...
- **SAFE** (ECB/EC): firms' own assessment of financing conditions, semiannual waves, **interview date recorded**
- **Estimation sample 2014m1–2023m9**: 57,647 observations, 1,981 firm–category pairs, 571 producer codes, 1,621 SAFE interviews

Who is in the matched sample

- Matching to scanner coverage selects **larger firms**: median assets about 8× the median SAFE–Orbis firm
- Much of the gap is sector composition: food and beverage manufacturers nearly half the sample, under 3% of SAFE
- Within-firm designs keep internal validity – external validity: established food-sector producers
- Survey-side inputs re-estimated on the full SAFE–Orbis panel, 138,500 firms: patterns reproduced

▸ Matched vs. full SAFE

Measuring prices: how the panel tracks the HICP

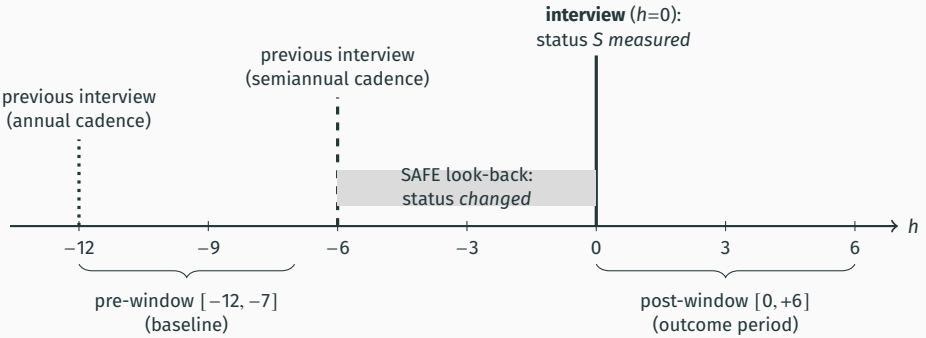


- Reference price: posted price, one-week sales removed, 13-week modal price, quantity-weighted monthly index per firm–product cell
- Reference prices change in **4.5%** of months, by **+5.2%** / **−5.0%** – posted-price benchmark 10.4%

▸ 12-month inflation

Empirical evidence

Timing: when the status is measured vs. changed



- Pre-window predates the look-back: clean baseline for the within-firm change
- New: credit terms at *measured* dates. Timing observed, not imputed

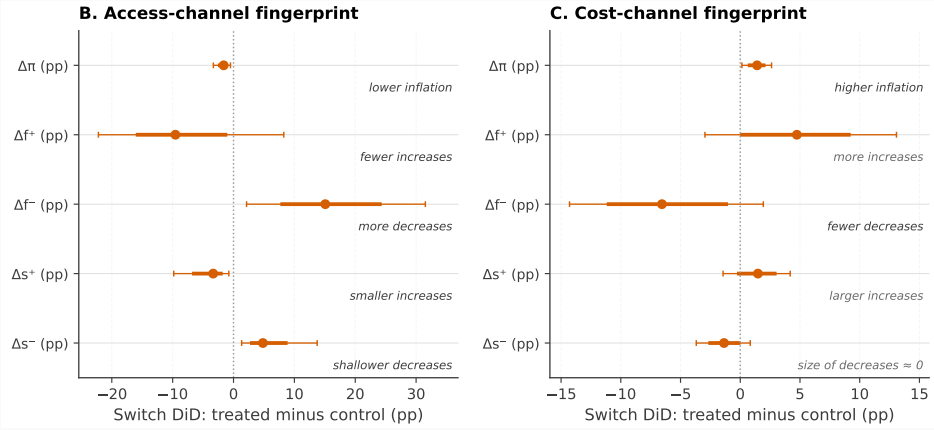
Design: interview-anchored within-firm comparisons

- Anchor: SAFE interview of producer v at month τ. Event time $h = t - \tau \in [-12, +6]$
- Building block: monthly log change π_{jt} of cell j's reference-price index, four margins:

$$f^{+} = \Pr(\pi > 0), \qquad f^{-} = \Pr(\pi < 0), \qquad s^{+} = \mathbb{E}[\pi \mid \pi > 0], \qquad s^{-} = \mathbb{E}[\pi \mid \pi < 0] < 0$$

- $\mathbb{E}[\pi] = f^{+}s^{+} + f^{-}s^{-}$ exactly. Since $s^{-} < 0$, a *positive* Δs^{-} means *shallower* cuts
- Within-interview change Δy: post-window $[0, +6]$ minus pre-window $[-12, -7]$
  All interviews: $\mathbb{E}[\Delta y \mid S{=}1] - \mathbb{E}[\Delta y \mid S{=}0]$
- **Switchers** (headline): firms whose credit status *changed* at the interview vs. firms that started in the same status and stayed

Headline: two dimensions of credit, opposite price responses



- Credit access worsened: inflation **falls**, –1.6pp/month. Credit cost worsened: inflation **rises**, +1.4pp/month
- Demand-side and profitability benchmarks show neither pattern

▸ Benchmarks

Five margins: what each channel does to price setting

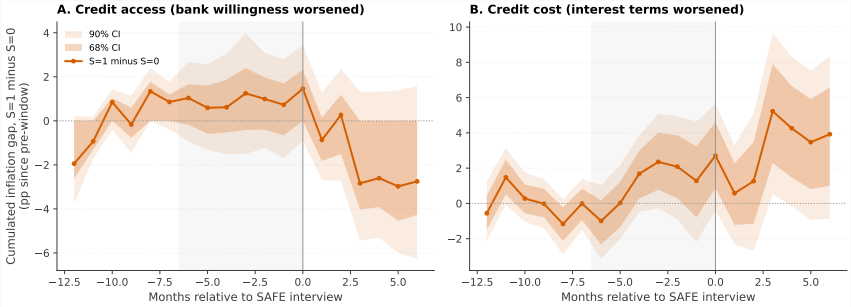
	Credit access worsened (bank willingness)		Credit cost worsened (interest-level terms)	
	Switchers	All interviews	Switchers	All interviews
$\Delta\pi$ (pp/month)	-1.6^{**}	-1.0^{**}	$+1.4^{*}$	-0.1
Δf^{+} (pp)	-9.5	-11.3^{**}	$+4.7$	-6.2
Δf^{-} (pp)	$+15.1^{*}$	$+3.4$	-6.6	-0.2
Δs^{+} (pp)	-3.3^{**}	-2.0^{*}	$+1.5$	$+0.2$
Δs^{-} (pp)	$+4.8^{**}$	$+3.0^{**}$	-1.4	-0.8
Treated / control events	21 / 191	28 / 409	23 / 78	59 / 179
Firm–product–months	2,840	6,042	1,239	3,395

Stars: the 90% (*), 95% (**) vendor-clustered bootstrap interval excludes zero.

- **Access**: increases rarer and compressed, cuts more frequent and shallower – defending cash flow *and* the customer base
- **Cost**: fewer cuts is the robust margin. All-interviews contrast: null

▸ Access indicators ▸ Cost measures ▸ Timing of cost switches

Dynamics around the interview



- **Access**: gap drifts *up* pre-interview, then reverses sharply. **Cost**: flat pre-window, rises only after the reported rate increase
- Mean reversion? Placebo six months earlier: +0.75pp – even full reversion explains under half of –1.6pp

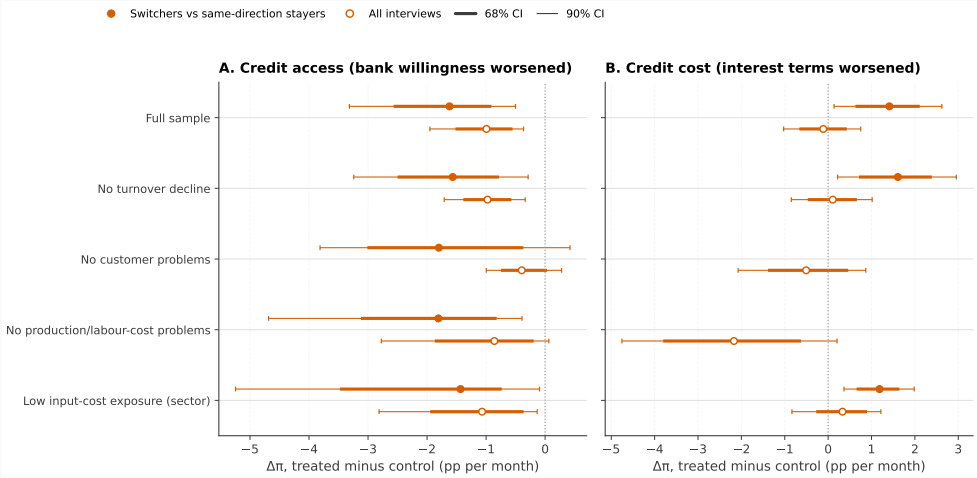
▸ Inference battery

Economic magnitudes

	Baseline level	Effect	Effect / baseline	Effect / SD
Credit access worsened				
Δf^{+} (pp)	11.4	−9.5	−83%	0.3
Δf^{-} (pp)	10.7	+15.1	+141%	0.5
Δs^{+} (pp)	5.2	−3.3	−64%	0.5
Δs^{-} (pp)	−5.6	+4.8	+86%	0.8
Credit cost worsened				
Δf^{+} (pp)	15.7	+4.7	+30%	0.2
Δf^{-} (pp)	13.8	−6.6	−48%	0.2
Δs^{+} (pp)	5.7	+1.5	+26%	0.2
Δs^{-} (pp)	−6.1	−1.4	−23%	0.2

- Channels **switch margins on and off**: access removes nearly all baseline increases, cost removes about half of baseline cutting
- Cumulated over seven post-months: transitory relative-price response $\approx -11\%$ (access), +10% (cost)

Financial, not relabelled demand or cost stress



- **Access**: negative everywhere, significant in every switcher cell (–1.4 to –1.8)
- **Cost** passes where populated: no turnover decline +1.6, low sectoral exposure +1.2

Where each channel lives: predetermined balance sheets

	Access events, by cash holdings		Cost events, by net leverage	
	Cash-poor	Cash-rich	Low leverage	High leverage
$\Delta\pi$ (pp/month)	-2.5^{*}	-1.4^{*}	$+2.3^{*}$	+0.6
90% band	[−4.7, −1.2]	[−5.3, −0.2]	[+0.5, +4.1]	[−1.2, +2.4]
Switch events	11	10	10	13
Observations	1,472	1,333	713	535

Switcher design, median split on Orbis exposure measured twelve months before the anchor interview.

- **Cost** concentrates in **low-leverage, low-debt-service** firms (+2.3 to +2.7pp), **access** strongest for **cash-poor** and most-levered firms – a mirror image
- Interpretation:
  - Firms with little debt can afford to pass higher interest rates on to customers
  - Firms short of cash cut prices to generate liquidity

▸ Which firms react

Quantities rule out a demand collapse

- **Why not demand?** If demand had collapsed, quantities would fall
- **Access**: prices down, quantities up – firms cut to sell more
- **Cost**: quantities do not fall – no demand collapse, but rising demand cannot be excluded

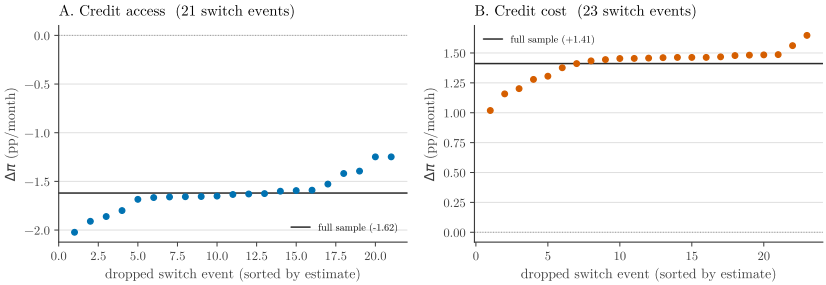
Switcher design	$\Delta\pi$ (pp/month)	$\Delta \log Q$ (%)
Credit access worsened	-1.6^{**}	$+33.4$
Credit cost worsened	$+1.4^{*}$	$+40.1^{*}$
Customer problems (demand benchmark)	-0.3	-46.4^{**}

Stars: the 90% (*), 95% (**) vendor-clustered bootstrap interval excludes zero.

- **Demand benchmark** as sanity check

▸ Demand benchmark figure

Robustness and inference: transparent about 21 and 23 events



- **Leave-one-event-out**: access $[-2.0, -1.3]$, cost $[+1.0, +1.7]$, sign preserved in every drop
- **More events**: bank-loan availability (22 switches) and the broad access-problem indicator (56 switches) reproduce the frequency anatomy

▸ Inference battery ▸ Access indicators ▸ Cost measures

Model

Environment: state-dependent pricing with working capital/cash

- **Pricing** (Costain–Nakov 2011): adjustment hazard increasing in the gain $D = M - V$, **symmetric** in direction
- **Working capital/cash**, Balleer et al. (2020): τ_w months' wage bill financed before revenue; external borrowing at net rate $\tilde{R}_k = R - 1 + \varsigma_k$
- **Financial state** $k \in \{\text{unconstrained, access-tight, cost-tight}\}$ follows a Markov chain

Flow payoff of a firm with customer base x:

$$u(q, x, s, k) = \underbrace{x(1+\phi_k)\mathcal{R}}_{\text{revenue + liquidity value}} - \underbrace{x\,\mathcal{W}}_{\text{wage bill}} - \underbrace{\tilde{R}_k\,\mathcal{E}}_{\text{financing cost}} + \underbrace{x\,\omega_k\,\bar{\pi}(k)}_{\text{customer premium}}$$

- Both channels reach the reset price through one composite wedge:

$$p^* = \tfrac{\varepsilon}{\varepsilon-1}\,\Omega_k\,ws, \qquad \Omega_k = \frac{1+\tau_w\tilde{R}_k}{1+\phi_k+\tilde{R}_k\,\xi_s} \qquad \text{access } \phi_k\uparrow \Rightarrow p^*\downarrow; \quad \text{cost } \varsigma_k\uparrow \Rightarrow p^*\uparrow$$

The customer base: cheap to lose, slow to regain

- **One-sided**: raising the price beyond a tolerance band $\bar{q}$ loses customers; cutting does not win them back
- **Large price changes cost customers**, more so when rationed (ω_k^c):

$$c(d) = a_c\,(1 + \omega_k^c)\,x\,|d|^{\eta}, \qquad \eta > 1, \qquad d = \text{price change},\ x = \text{customer base}$$

$\Rightarrow$ Firms close price gaps in **small steps**: increases rare and small, cuts frequent but shallow

$\Rightarrow$ **The customer base sets the step size**

▸ Full customer block ▸ Why this band

Two analytical results

Proposition 1: signs of the two channels

- Tighter access lowers the reset price: $\partial p^* / \partial \phi_k < 0$
- Expensive credit raises it: $\partial p^* / \partial \varsigma_k > 0$ iff $\xi_s < \tau_w(1 + \phi_k)$ – holds at $\tau_w = 3.2$ for all $\xi_s \in [0, 1]$

Proposition 2: identifying the mechanism

- If the state only shifts the target down and adjusters jump to it, cuts get *larger*
- Data show the **forbidden pair**, more frequent *and smaller* cuts $\Rightarrow$ the state must enter the adjustment technology

- **Frequency margins identify the level of adjustment frictions, size margins their shape**

Taking the model to the data: what disciplines what

- **Three financial states from SAFE**: transition matrix from 142 consecutive interview pairs
- Stationary shares 73/8/19%, monthly staying probabilities 0.87 (access) and 0.89 (cost)
- **Calibrated**: $\varepsilon = 7$, hazard shape from Costain–Nakov, $\tau_w = 3.2$ months = median Orbis working capital over monthly operating costs
- **Pricing block fitted to the panel's own reference prices** (product–store quote): frequency 4.5% and sizes +5.2/−5.0% pin down $\bar{\lambda}$, σ_{MC}, a_c; the channel fingerprints are held out (preliminary, 9 Nov)
- **September draft**: SMM of 9 parameters incl. both channels' *frequency* fingerprints in matched units (fit table next)
- **Model and data measured identically**: simulated monthly panel, semiannual "interviews", same switcher estimator and clustered bootstrap

▸ Financial-state process ▸ Calibration and SMM ▸ Re-calibration

Fit: the ten channel fingerprints

	Model		Data (switchers)		
	firm units	cell units	estimate	90% band	in band
Credit access worsened					
$\Delta\pi$ (pp/m)	−0.13	−0.14	−1.6	[−3.3, −0.5]	–
Δf^{+} (pp)	−1.10	−2.09	−9.5	[−22.2, +8.3]	✓
Δf^{-} (pp)	+1.65	+2.56	+15.1	[+2.1, +31.5]	–
Δs^{+} (pp)	−1.87	−0.07	−3.3	[−9.8, −0.8]	✓
Δs^{-} (pp)	+0.53	+0.06	+4.8	[+1.3, +13.7]	–
Credit cost worsened					
$\Delta\pi$ (pp/m)	+0.07	+0.06	+1.4	[+0.1, +2.6]	–
Δf^{+} (pp)	+0.68	+1.75	+4.7	[−3.0, +13.1]	✓
Δf^{-} (pp)	−0.52	−1.18	−6.6	[−14.3, +1.9]	✓
Δs^{+} (pp)	+0.60	−0.04	+1.5	[−1.4, +4.2]	✓
Δs^{-} (pp)	+0.58	−0.06	−1.4	[−3.7, +0.8]	✓

- **9 of 10** signs match, **6 of 10** in band. The four outside: right-signed but smaller

▸ Matched units

The access-side inflation gap: rotation, not deflation

- Model −0.13pp vs. headline −1.62: a **rotation** – fewer, smaller increases offset by shallower cuts

	Switchers		All interviews	
Indicator	$\Delta\pi$	90% band	$\Delta\pi$	90% band
Bank willingness (headline)	−1.62	[−3.3, −0.5]	−0.99	[−2.0, −0.4]
Bank-loan availability	−0.23	[−1.6, +1.2]	−0.70	[−1.7, +0.1]
Access a pressing problem (5× the events)	+0.05	[−0.8, +0.7]	−0.32	[−0.7, +0.1]

- The better measured the access treatment, the smaller the response: a **reallocation of adjustment**, not deflation

▸ Access indicators

Mechanism decomposition: each block required by a different margin

Access-tight firms face	Δf^+	Δf^-	Δs^+	Δs^-
Rationing, margin financed 0.5 months (no customer base)	−0.23	+0.41	−0.27	−0.40
+ convex repricing cost with premium ω^c	−0.23	+0.41	−1.55	+0.64
+ one-sided retention, no flow premium	+0.24	+0.14	−3.02	−0.15
+ flow premium ω (full block)	−0.55	+0.44	−1.33	+0.54
Data (cell units)	−10.89	+12.95	−3.40	+4.81

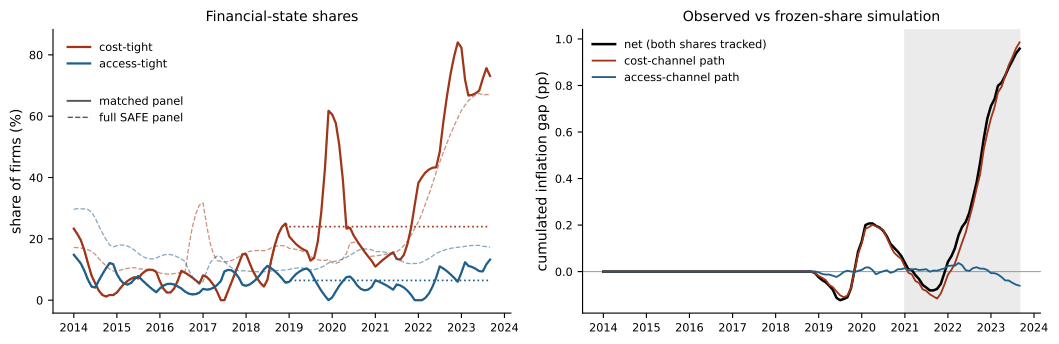
Tight minus slack, pp. Own-change band at the draft's parameters, 31-point grid, wage 0.882 (row 1 general equilibrium, rows 2–4 partial equilibrium at that wage; the two coincide for this band). Model rows firm-level, data row cell-level: compare directions, not levels.

- Rationing with a short financing margin sets the **direction**, cutting, but makes cuts **deeper** (underlined) – the target-shift pattern the data reject
- The repricing premium ω^c flips cuts to **shallower**; retention alone only deters increases; the flow premium ω turns it into the full access pattern

Aggregate application: the 2021–23 surge

Did credit conditions amplify the surge? A composition accounting

- Observed vs 2019-frozen SAFE tight-state shares, fed through estimated policies



- Net +0.96pp of 2021–23 inflation in the covered basket: cost +0.99, access −0.06

▸ Full-panel state paths ▸ Calvo nest

Conclusion

- **Two dimensions of credit, opposite signs**. Less willing banks ⇒ prices fall: fewer, smaller increases – Kim 2021
- Expensive credit ⇒ prices rise: fewer cuts – Gilchrist et al. 2017
- The **size margins** identify the mechanism: more frequent *and shallower* cuts reject every target-shift model
- Access **rotates** adjustment, cost moves the price level: 2021–23 adds ≈ +1pp to the covered basket, essentially all via cost
- **Ongoing**: endogenizing the financial state, better fit

Backup

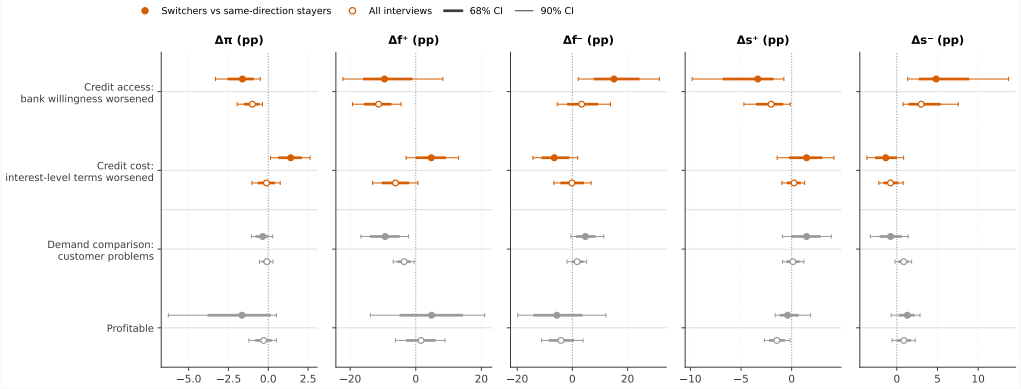
Backup: 12-month inflation in the panel



- Panel inflation is subdued until 2021, then rises sharply, mirroring euro-area food and household-product HICP inflation

Back

Backup: the channels against the demand and profitability benchmarks



- Same design and firms: customer problems and profitability show neither credit fingerprint

◂ Back: Headline ◂ Back: Quantities

Backup: related literature

- **Financial conditions and price setting**: Gilchrist–Schoenle–Sim–Zakrajšek 2017, Balleer et al. 2020, Renkin–Züllig 2024, Bauer et al. 2026
- **Cost channel**: Barth–Ramey 2001, Christiano et al. 2005, Gaiotti–Secchi 2006
- **Customer markets**: Phelps–Winter 1970, Gottfries 1991, Chevalier–Scharfstein 1996, Gourio–Rudanko 2014
- **Micro price setting**: Nakamura–Steinsson 2008, Eichenbaum–Jaimovich–Rebelo 2011, Kehoe–Midrigan 2015, Gautier et al. 2024
- **Financial frictions and real outcomes**: Bernanke–Gertler–Gilchrist 1999, Gilchrist–Zakrajšek 2012, Ottonello–Winberry 2020
- **Modeling**: Costain–Nakov 2011, Reiter 2009, Klein 2000

Back

Backup: a Calvo nest – the composition-augmented Phillips curve

- Calvo with survival θ in every financial state. ω_t: constrained share, Ω: its reset-price wedge

$$\pi_t = \beta\,\mathbb{E}_t\pi_{t+1} + \kappa\,\widehat{mc}_t + \kappa\log\Omega\cdot\omega_t, \qquad \kappa = \frac{(1-\theta)(1-\beta\theta)}{\theta}$$

- Composition enters like a **cost-push term**: $\log\Omega < 0$ for access, $\log\Omega > 0$ for cost (Proposition 1)
- Tighter access is a negative cost-push shock, expensive credit a positive one – the analytic counterpart of the surge accounting
- With $\theta_c \neq \theta_u$ composition also moves the *slope*, $\bar{\kappa}(\omega) = (1-\omega)\kappa_u + \omega\kappa_c$ – small at $\omega \approx 8\%$
- Calvo is direction-blind: it carries the target shift, not the frequency-and-size anatomy (Proposition 2) – hence the quantitative model

Back

Backup: corroborating access indicators

	Bank willingness	Loan availability	Pressing problem ($\geq 7/10$)
Switchers vs. same-direction stayers			
$\Delta\pi$ (pp/m)	-1.6^{**}	-0.2	$+0.1$
Δf^{+} (pp)	-9.5	-3.9	-6.6^{*}
Δf^{-} (pp)	$+15.1^{*}$	$+11.6^{**}$	$+3.8$
Δs^{+} (pp)	-3.3^{**}	$+1.9$	$+0.3$
Δs^{-} (pp)	$+4.8^{**}$	$+1.1$	$+0.6$
Switch events	21	22	56
All interviews			
$\Delta\pi$ (pp/m)	-1.0^{**}	-0.7	-0.3
Δf^{+} (pp)	-11.3^{**}	-8.0^{**}	-3.4
Δf^{-} (pp)	$+3.4$	$+2.1$	$+4.2^{*}$
Treated events	28	33	153

- Frequency anatomy reproduced by the availability item and the broad pressing-problem rating
- Inflation response specific to the sharpest treatment, bank willingness

◂ Back: Five margins ◂ Back: Robustness ◂ Back: Access-side gap

Backup: corroborating cost measures (switcher design)

	Interest level	Other fin. costs[a]	Input costs	Labour costs
$\Delta\pi$ (pp/m)	+1.4*	+0.9	+0.2	−0.1
Δf^{+} (pp)	+4.7	+1.2	−2.3	+1.4
Δf^{-} (pp)	−6.6	−13.9**	−1.9	−1.7
Δs^{+} (pp)	+1.5	+2.7**	−0.3	−0.6
Δs^{-} (pp)	−1.4	+0.3	+1.9**	+0.8
Switch events	23	16	76	77

[a] Charges, fees, commissions. Low-leverage arm of the exposure split only.

- The contract's second price term reproduces the cost anatomy: fewer cuts, larger increases
- Input and labour cost increases also suppress the downward margin – remaining cuts shallower

◂ Back: Five margins ◂ Back: Robustness

Backup: inference battery and design robustness ($\Delta\pi$, switcher design)

	Credit access (willingness)	Credit cost (interest level)
Baseline (winsor 2/98)	-1.62 $[-3.32, -0.50]$	$+1.41$ $[+0.14, +2.62]$
bootstrap p (two-sided)	0.018	0.072
Leave-one-event-out range	$[-2.02, -1.25]$	$[+1.02, +1.65]$
Tightening only (0→1 vs. 0→0)	-1.89 $[-4.18, -0.58]$	$+0.90$ $[-0.32, +1.99]$
Easing only (1→0 vs. 1→1, unflipped)	$+1.27$ $[-1.30, +3.82]$	-2.20 $[-4.11, -0.10]$
Randomization p: event-level / vendor-block	0.266 / 0.020	0.025 / 0.040
Semiannual cadence only (15 / 16 events)	-2.09 $[-3.58, -0.90]$	$+1.72$ $[+0.21, +3.38]$
No next interview inside post-window	-1.21 $[-2.83, +0.01]$	$+1.16$ $[-0.46, +2.68]$
Category × month demeaned π	-2.53 $[-7.52, -0.35]$	$+1.46$ $[+0.11, +2.76]$
Winsor (1,99)	-1.65 $[-3.48, -0.48]$	$+2.16$ $[-0.14, +4.72]$
No winsorization	-0.89 $[-2.72, +0.75]$	$+2.94$ $[-0.61, +7.21]$
Placebo: windows 6 months earlier	$+0.75$ $[+0.0, +1.7]$	$+0.3$ $[-0.7, +1.3]$
Alternative pre-windows	-1.0 to -1.9	–
Stacked event study	-1.8 ($p = 0.03$)	–

◂ Back: Dynamics ◂ Back: Robustness

Backup: which firms react (switcher $\Delta\pi$, predetermined characteristics)

	Credit access worsened		Credit cost worsened	
	$\Delta\pi$	Events	$\Delta\pi$	Events
Markup: low	-2.6^*	6	-2.0^*	5
Markup: high	-1.6^*	4	$+0.8^*$	6
Size: small	-2.0^*	11	$+2.1$	11
Size: large	-2.5^*	7	-0.4	8
Age: younger	-1.2	11	$+1.7$	13
Age: older	-1.8^*	10	$+0.8$	10

- Access response: present in every cell
- Cost response: concentrated in high-markup and smaller firms
- Predetermined balance-sheet and inventory controls leave the access estimate at -1.5
- Cost estimate on the controls subsample: -0.6, n.s., 12 events – sample composition, not covariates
- Four to thirteen switch events per cell: suggestive

Back

Backup: the financial-state process

- **Mostly idiosyncratic**: wave × country factors explain 17–34% of interview-level variance for access indicators, 58% for the interest level
- Permanent firm heterogeneity: 38–54% for access, 27% for the interest level
- ⇒ credit access is a firm-specific state, credit cost co-moves with the rate cycle
- **Persistent**: first-order Markov chains fit the wave-to-wave transitions, monthly staying probabilities 0.85 to 0.94
- **Cadence bound**: six-month-gap pairs imply 0.88 monthly for access, the value the model uses
- **Adjustment hazard**: 0.90 one month into a price spell, 0.17 after six months – a signature Calvo cannot reproduce

Back

Backup: calibration and the September draft's SMM estimates

Calibrated (monthly)		
Discount factor	β	$1.04^{-1/12}$
Risk aversion	γ	2
CES elasticity	ε	7
Hazard shape	α, ζ	0.037, 0.235
Marginal-cost s.d.	σ_{MC}	0.155
Financing duration	τ_w	3.2
Collateral, pledgeability	ξ_c, ξ_s	0, 0.5
Baseline attrition	δ_0	0.05
Tolerance band	$\bar{q}, \sigma_q$	0.04, 0.02
Transition matrix	F	measured

Estimated by SMM			
Adjustment rate	$\bar{\lambda}$	0.0408	(0.0117)
Cost persistence	ρ_s	0.9903	(0.0089)
Liquidity value, access	ϕ_2	0.116	(0.242)
Flow premium, access	ω_2	5.04	(13.26)
Repricing premium, access	ω_2^c	16.58	(188.54)
Repricing cost scale	a_c	0.200	(2.442)
Repricing curvature	η	2.96	(14.88)
Price-sensitive attrition	δ_1	0.183	(0.378)
Flow premium, cost	ω_3	1.09	(5.66)
Step window (grid months)	m	16	

- Repricing curvature interior: $\eta \approx 3$
- Cost state's customer premium active: $\omega_3 \approx 1.1$
- (a_c, ω_2^c, η) jointly, not separately, identified: correlations up to 0.98
- Re-solving at $\tau_w \in \{1.6, 4.8\}$: access fingerprints unchanged, cost fingerprints scale with $\varsigma_3 \tau_w$

▸ Re-calibration (9 Nov) ◂ Back

Backup: the customer block in full

- Base x shifts demand out at any relative price and reacts to the **firm's own price change** $d_t = \log p_t - \log p_{t-1}$ (0 without adjustment):

$$x_{t+1} = \varrho(d_t)\, x_t + \delta_0, \qquad \varrho(d) = 1 - \delta_0 - \delta_1 \left[\Lambda\left(\tfrac{d-\bar{q}}{\sigma_q}\right) - \Lambda\left(\tfrac{-\bar{q}}{\sigma_q}\right) \right]^+$$

- **One-sided**: attrition rises only when the firm raises its price by more than the tolerance band $\bar{q}$; cuts and unchanged prices keep retention at $1 - \delta_0$ – cuts do not buy customers back
- **Repricing is costly in customers**: an adjusting firm chooses a step d, not a jump to the target

$$M = \max_d \{V(q+d, x, s, k) - c(d; x, k)\}, \qquad c(d; x, k) = a_c\,(1 + \omega_k^c)\, x\, |d|^\eta, \quad \eta > 1$$

- Convex repricing cost $\Rightarrow$ gaps close in small steps: runs of same-signed changes
- No direction imposed: hazard, wedges, repricing cost symmetric – increase/decrease asymmetry is an **equilibrium outcome** of one-sided retention

Back

Backup: where the tolerance band sits decides what the model can do

Band on the price	relative to market level	relative to own flexible price	own change d (previous price) draft logistic	floored†
Frequency of changes	6.1%	9.7%	7.0%	5.5%
Kurtosis	1.8	2.6	2.7	2.5
Changes $\leq$ 5pp	0%	31%	41%	36%
Firms at the minimum base	23.5%	6.2%	0.0%	0.0%
Δf^{+} (pp)	−0.49	−0.22	−0.53	−0.42
Δf^{-} (pp)	+0.28	+0.65	+0.44	+0.29
Δs^{+} (pp)	−5.15	−0.88	−1.55	−2.18
Δs^{-} (pp)	+1.63	−0.15	+0.52	+0.23
Δ mean log price	−1.46%	−0.97%	−0.85%	−0.84%
	cost sorting	pair lost	both fingerprints	both, no cut bonus

Full customer block at the draft's parameters, two states (8% access-tight), 31-point grid, tight minus slack; GE wages 0.960, 0.927, 0.882. †PE at the GE wage; PE and GE coincide for this band.

- A market-level band sorts firms by permanent cost, an own-level band loses the shallower cuts; the band on the firm's own change needs no extra state and gives a plausible economy *and* both fingerprints

Back

Backup: re-calibration on the panel's reference-price moments (preliminary)

			fitted, ρ_S =	
Monthly	data	start	0.90	0.99
Frequency of changes	4.5%	6.6%	4.5%	4.5%
Size of increases	5.2%	9.7%	4.9%	4.8%
Size of decreases	5.0%	8.9%	5.6%	5.5%
Share of decreases (not targeted)	36%	52%	47%	46%
Adjustment rate $\bar{\lambda}$		0.110	0.082	0.086
Marginal-cost s.d. σ_{MC}		0.155	0.116	0.123
Repricing cost scale a_c		0.20	0.18	0.22

			fitted, ρ_S =	
Held out, tight − slack	data	start	0.90	0.99‡
Δf^+ (pp)	−3.2	−0.5	−0.5	−0.4
Δf^- (pp)	+5.5	+0.3	+0.3	+0.4
Δs^+ (pp)	−3.4	−1.6	−0.6	−0.7
Δs^- (pp)	+4.8	+0.3	−0.1	+0.2
Δ mean log price	−1.7%	−0.7%	−0.4%	−0.4%

Data: switcher estimates scaled to the single-product unit. 0.90: general-equilibrium re-solve (wage 0.865). ‡Partial equilibrium at the GE wage; for this band PE and GE agree to 0.02pp.

- Frequency matched; the sizes **cross**: the band compresses increases, nothing compresses cuts
- Fingerprints an order of magnitude below the data and the cut-size gap vanishes: the gap sizes rest on the **financial block**, which the pricing moments do not touch
- Next: free the tight-state rationing and the premia on the four access fingerprints

Back

Backup: matched-unit measurement and state-path representativeness

- **Matched units**: model panels pass through the data's measurement pipeline – staggered store adoption, index construction, cell aggregation
- Aggregation attenuates size margins mechanically, so the size fingerprints are scored at the firm level
- **State paths**: matched sample ≈ 23 interviews a month vs. ≈ 6,300 in the full SAFE–Orbis panel
- Cost-tight share: correlation 0.88, surge-window averages 53.8 vs. 52.1%
- Access-tight share: 14.5% full panel vs. 6.4% matched – established food producers are least likely to report less willing banks
- Counterfactual with full-panel paths: +0.96 → +0.88pp, cost +1.02, access −0.18

◂ Back: Sample ◂ Back: Fit ◂ Back: Surge

Backup: the timing of the cost switches and the credit-outcome taxonomy

- Rate increases: uncommon before 2022, widespread in the 2022–23 hiking cycle
- The *treated share* concentrates in 2022–23, hence the null all-interviews contrast
- Only about a quarter of the 23 *switch events* anchor there: identification from firm-specific rate-increase episodes over 2015–2023
- Rate question answered only by credit applicants, so switchers are more likely active borrowers
- Effect concentrated in low-leverage firms, absent among the most credit-engaged: hard to reconcile with borrower selection driving the result

Credit outcome (share of firm-months)	No tightening	Cost only	Access only	Both
Financially constrained[1]	6.6%	6.3%	23.9%	31.5%
of which: application rejected	2.4%	0.0%	8.0%	11.3%
Rejection or discouragement		2.3%	11.1%	

[1] SAFE definition: application rejected, only a limited amount granted, offer declined because the cost was too high, or did not apply for fear of rejection.

- Motivates the model's hierarchical state assignment: joint reporters → access-tight

◂ Back: Credit conditions ◂ Back: Five margins